



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Continuous Assessment Test-II, Winter 2019-2020

CSE2003- Data Structures and Algorithms

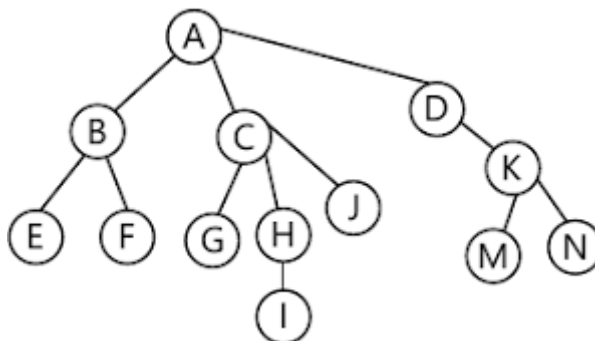
Slot: B2

Exam Duration: 90 Minutes

Maximum Marks: 50

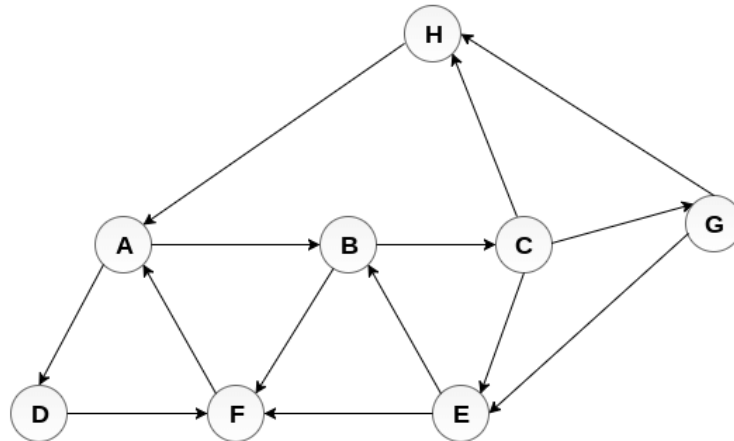
Answer all questions (5*10 = 50)

1. Define a structure to accept name and age of N employees to store in a single linked list. Engrave code to create two lists in which one list has age of type odd number and other list has age of type prime number. Display both the lists.
2. Mention the rule to convert general tree to unique binary tree and convert the given general tree to unique binary tree. On a converted binary tree perform preorder traversal, inorder traversal and postorder traversal.



3. What is the importance of hashing? Given input 4371, 1323, 6173, 4199, 4344, 9679, 1989 and apply a hash function $h(x) = (x) \bmod 11$. If collisions occur, resolve using following methods.
 - a. Separate chaining method
 - b. Quadratic probing method
4. a) Engrave a pseudo code to input N elements and perform the following (5)
 - i. Insert into a heap one by one.
 - ii. Build max heap.

b) Consider a graph $G = (V, E)$ and perform DFS (G). Provide output of a graph with its starting vertex as A and mark the vertices in the order they visited. Show the vertex selection trace. (5)



5. Consider a weighted directed graph $G = (V, E)$ and find shortest path from vertex b to all other vertices using Dijkstra's algorithm. Show the trace table at each phase and inscribe the algorithm also.

